

RESEARCH PAPER

Anomaly Detection in IoT Networks Using Federated Machine Learning

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ABSTRACT

The proliferation of Internet of Things devices has introduced unprecedented security challenges in modern network infrastructures. This paper proposes a federated learning-based anomaly detection framework for identifying malicious traffic patterns in heterogeneous IoT environments. Our approach enables distributed model training across edge nodes without centralizing sensitive network data, thereby preserving privacy while maintaining detection accuracy. Experimental evaluation on the CIC-IoT-2023 and N-BaloT benchmark datasets demonstrates that our federated approach achieves 96.2 percent detection accuracy with a false positive rate below 1.8 percent, comparable to centralized baselines while reducing data transmission overhead by 78 percent.

Keywords: IoT Security, Federated Learning, Anomaly Detection, Network Intrusion Detection, Edge Computing

1. INTRODUCTION

The rapid expansion of Internet of Things ecosystems has created vast attack surfaces that traditional perimeter-based security models cannot adequately protect. By 2025, an estimated 75 billion IoT devices are projected to be deployed globally, generating massive volumes of heterogeneous network traffic. Centralized monitoring approaches face scalability limitations and raise significant privacy concerns when processing data from distributed sensor networks. Federated learning offers a promising paradigm for collaborative model training that keeps raw data at the network edge.

2. SYSTEM ARCHITECTURE

Our proposed framework consists of three principal components: (1) lightweight traffic feature extractors deployed on IoT gateway nodes, (2) local anomaly detection models trained using gradient-based federated optimization, and (3) a central aggregation server that coordinates model updates using secure aggregation protocols. Each edge node maintains a local Random Forest and LSTM hybrid model that processes network flow statistics including packet rates, byte distributions, and protocol entropy measurements.

3. EXPERIMENTAL EVALUATION

We evaluate our framework using two established IoT security benchmarks. The CIC-IoT-2023 dataset contains 33 attack categories across smart home, industrial, and healthcare IoT scenarios. The N-BaloT dataset comprises traffic from nine commercial IoT devices under both benign and botnet-infected conditions. Our federated approach achieves an F1-score of 0.954 on CIC-IoT-2023 and 0.968 on N-BaloT, with convergence achieved within 15 communication rounds.